

KUUSKAJASKARI, Finland

WMO: 029610

Lat: 61.133N

Lon: 21.366E

Elev: 10

StdP: 101.20

Time Zone: 2.00 (EUE)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.5	-16.2	-22.3	0.5	-19.6	-18.7	0.7	-15.9	18.8	3.4	16.3	1.7	3.0	50	0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.5	22.9	18.0	21.3	17.2	20.0	16.5	18.8	21.9	17.8	20.6	16.8	19.2	4.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	12.5	20.5	16.5	11.7	19.5	15.6	11.1	18.5	53.3	21.9	50.0	20.7	47.2	19.2	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	13.4	11.8	DB	-19.3	25.2	6.0	1.9	-23.6	26.6	-27.2	27.7	-30.6	28.7	-35.0	30.1	(4)
(5)			WB	-19.7	20.0	5.3	1.6	-23.6	21.2	-26.7	22.1	-29.7	23.0	-33.6	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.4	-3.4	-4.5	-1.0	3.4	8.2	13.2	16.0	15.4	11.1	6.5	1.3	-1.8	(6)
(7)		DBStd	8.06	5.94	5.84	3.75	3.22	3.03	2.71	2.29	2.73	2.82	3.28	3.85	4.69	(7)
(8)		HDD10.0	2197	416	406	341	199	73	4	0	1	19	111	262	366	(8)
(9)		HDD18.3	4727	674	639	599	448	313	157	80	97	217	366	512	624	(9)
(10)		CDD10.0	528	0	0	0	2	18	98	185	169	52	4	0	0	(10)
(11)		CDD18.3	16	0	0	0	0	0	2	7	7	0	0	0	0	(11)
(12)		CDH23.3	31	0	0	0	0	0	4	16	11	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.5	6.5	5.9	5.5	5.0	4.7	4.3	4.7	4.8	5.3	6.3	6.0	6.6	(14)
(15)	Precipitation	PrecAvg	630	51	37	33	32	38	56	68	71	58	69	60	58	(15)
(16)		PrecMax	818	100	78	58	61	85	108	134	137	128	159	139	108	(16)
(17)		PrecMin	514	7	2	5	1	5	18	2	25	12	7	9	7	(17)
(18)		PrecStd	77	23	20	13	16	20	23	33	32	35	35	28	26	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	4.7	5.9	15.1	20.2	23.7	25.6	24.7	19.7	13.5	8.7	5.5	(19)
(20)			MCWB	2.9	3.9	4.0	10.2	14.3	18.1	20.0	19.4	15.8	12.1	7.7	4.5	(20)
(21)		2%	DB	3.1	3.3	4.6	12.1	17.4	21.3	22.9	22.8	17.9	12.1	8.1	4.7	(21)
(22)			MCWB	2.2	2.5	3.1	8.6	12.9	16.6	18.0	18.0	14.7	11.0	7.2	3.9	(22)
(23)		5%	DB	2.6	2.5	3.6	10.0	15.0	19.3	21.5	21.1	16.3	11.3	7.1	4.0	(23)
(24)			MCWB	1.7	1.8	2.4	7.1	11.3	15.4	17.3	17.6	14.0	10.2	6.2	3.3	(24)
(25)		10%	DB	1.9	1.7	2.8	8.2	13.1	17.7	20.1	19.6	15.0	10.5	6.1	3.3	(25)
(26)			MCWB	1.2	1.1	1.8	6.0	9.9	14.6	16.6	16.7	13.2	9.5	5.2	2.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	4.0	4.4	10.6	15.1	18.9	21.1	20.0	16.6	12.5	8.1	4.9	(27)
(28)			MCDWB	3.5	4.5	5.3	14.5	19.1	22.6	25.0	23.3	18.6	13.1	8.7	5.3	(28)
(29)		2%	WB	2.5	2.7	3.3	8.8	13.3	17.3	18.8	18.8	15.5	11.2	7.3	4.1	(29)
(30)			MCDWB	2.9	3.2	4.1	11.8	17.0	20.4	21.6	22.0	17.3	11.9	7.9	4.5	(30)
(31)		5%	WB	1.9	1.9	2.6	7.4	11.6	15.9	17.7	17.8	14.4	10.4	6.3	3.4	(31)
(32)			MCDWB	2.4	2.4	3.3	9.7	14.8	18.6	20.7	20.5	16.0	11.1	6.9	3.9	(32)
(33)		10%	WB	1.3	1.2	1.9	6.1	10.0	14.9	16.7	17.0	13.3	9.6	5.3	2.7	(33)
(34)			MCDWB	1.8	1.6	2.6	8.0	12.6	17.2	19.4	19.1	14.8	10.3	6.0	3.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.9	4.3	4.2	5.0	5.9	5.5	4.9	4.3	3.0	3.0	3.9	(35)	
(36)			MCDBR	3.3	2.8	4.3	8.2	8.7	8.1	7.9	7.1	5.7	3.4	2.6	3.0	(36)
(37)			MCWBR	3.2	2.5	3.2	5.1	5.0	4.4	4.1	3.7	3.4	2.9	2.8	2.8	(37)
(38)		5% WB	MCDBR	3.0	2.7	3.6	7.9	8.3	6.7	7.3	6.4	5.0	3.2	2.6	2.9	(38)
(39)			MCWBR	3.1	2.6	2.8	5.1	5.0	4.2	4.2	3.7	3.2	2.9	2.7	2.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.311	0.329	0.368	0.366	0.357	0.385	0.372	0.346	0.329	0.290	0.362	(40)	
(41)		taud	2.151	2.246	2.317	2.323	2.412	2.473	2.422	2.446	2.500	2.398	2.197	2.554	(41)	
(42)		Ebn at Noon	453	661	789	823	857	872	834	815	774	643	417	299	(42)	
(43)		Edh at Noon	38	66	86	104	102	98	101	91	73	57	37	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.20	0.78	2.32	3.96	5.53	6.09	5.61	4.37	2.61	1.08	0.28	0.12	(44)	
(45)		RadStd	0.03	0.15	0.26	0.39	0.45	0.37	0.49	0.38	0.28	0.17	0.05	0.01	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (48 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-120	N/A	N/A	

Nomenclature: See separate page